

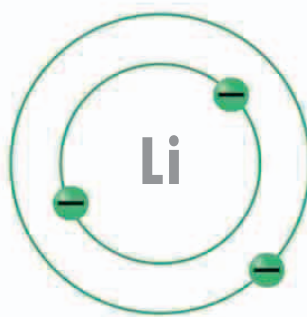
Electron configurations

As the atomic number increases, atoms get heavier and larger, because the electrons are arranged in shells positioned farther and farther out from the nucleus. The first shell can hold two electrons and the second can hold eight. Once the third shell has eight, the fourth shell starts to fill up, although in some cases these shells can hold many more than eight electrons (see page 124).

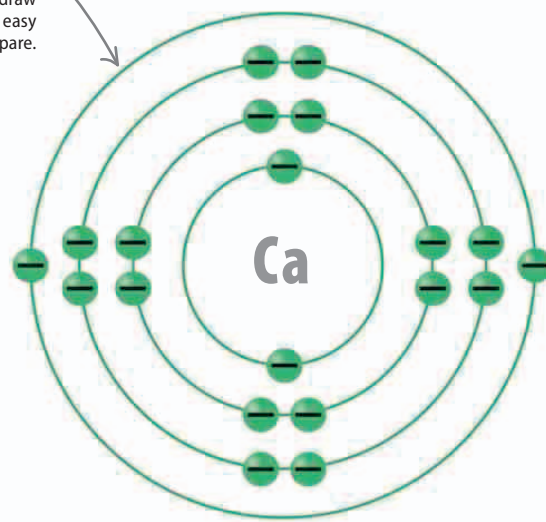
Shell shape
In reality the electron shells are not round. Scientists draw them like this so they are easy to see and compare.



² He **△ Helium**
With an atomic number of 2, helium has two protons and therefore two electrons in a single shell.



³ Li **△ Lithium**
Lithium has an atomic number of 3. The first electron shell is full, so the third electron sits in another shell.



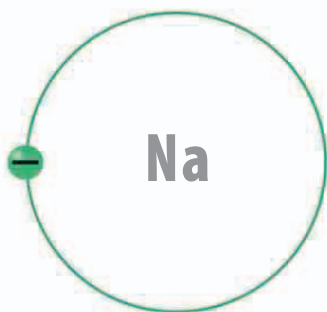
²⁰ Ca **△ Calcium**
With an atomic number of 20, the electrons in a calcium atom are arranged over four shells.

Outer shell

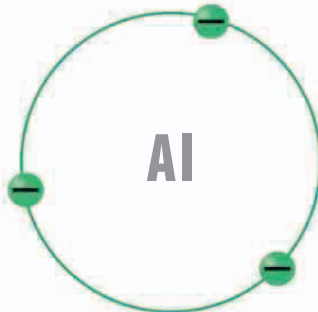
The electrons in an atom's outer shell are the ones that form bonds with other atoms and become involved in chemical reactions. So the number of outer electrons in an atom is a strong indicator of an element's physical and chemical properties. Atoms react with each other to achieve a full outer shell and therefore become more stable. The diagrams below show only the outer shell of each atom.

▽ Octet rule

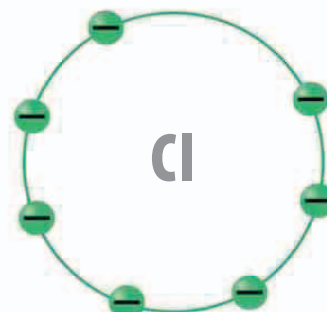
Atoms need to have eight electrons in their outer shell to become stable. This is called the octet rule. They must either gain electrons to reach eight, or lose electrons so that the next shell down—which will be full—becomes their outer shell.



¹¹ Na **△ Sodium**
A sodium atom has just one outer electron. To get a full outer shell, it must give that electron away.



¹³ Al **△ Aluminum**
An aluminum atom has three outer electrons. It has to lose all of these to become stable.



¹⁷ Cl **△ Chlorine**
A chlorine atom has seven outer electrons. It has space for one more, which would fill its outer shell.